

Cardioexaban for Secondary Prevention in High-Risk ASCVD

The SHIELD-2 Confirmatory Trial

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ABSTRACT

Background

The SHIELD-1 trial demonstrated that cardioexaban, a dual P2Y₁₂ and PAR-1 inhibitor, reduced major adverse cardiovascular and cerebrovascular events (MACCE) in high-risk patients with atherosclerotic cardiovascular disease (ASCVD). SHIELD-2 was designed as a confirmatory trial to validate these findings in an independent population.

Methods

In this multicenter, randomized, double-blind, placebo-controlled trial, we enrolled 7,842 patients with documented ASCVD and elevated residual cardiovascular risk (10-year ASCVD risk score $\geq 15\%$ or ≥ 2 high-risk features) despite guideline-directed medical therapy. Patients were randomly assigned 1:1 to cardioexaban 150 mg once daily or placebo. The primary endpoint was time to first occurrence of MACCE (cardiovascular death, myocardial infarction, stroke, or urgent coronary revascularization). Key secondary endpoints included three-point MACE (cardiovascular death, MI, or stroke), individual MACCE components, and stroke specifically.

Results

Over a median follow-up of 36 months, MACCE occurred in 449 patients (11.5%) in the cardioexaban group versus 590 patients (15.1%) in the placebo group (hazard ratio, 0.73; 95% confidence interval [CI], 0.63 to 0.85; $P < 0.001$). Three-point MACE occurred in 372 patients (9.5%) versus 503 patients (12.9%) (HR, 0.71; 95% CI, 0.59 to 0.85; $P < 0.001$). Stroke occurred in 96 patients (2.5%) versus 142 patients (3.6%) (HR, 0.68; 95% CI, 0.51 to 0.89; $P = 0.006$). TIMI major bleeding occurred in 160 patients (4.1%) versus 90 patients (2.3%) (HR, 1.77; 95% CI, 1.35 to 2.33; $P < 0.001$), predominantly gastrointestinal. Intracranial hemorrhage rates were balanced (0.4% vs. 0.3%).

Conclusions

SHIELD-2 confirmed the findings of SHIELD-1, demonstrating that cardioexaban reduces MACCE in high-risk ASCVD patients on guideline-directed therapy, with particular benefit on stroke prevention. Major bleeding was increased but driven primarily by manageable gastrointestinal events.

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1 INTRODUCTION

The SHIELD-1 trial established that cardioexaban—a novel dual-pathway antiplatelet agent inhibiting both P2Y₁₂ and protease-activated receptor-1 (PAR-1)—reduces major adverse cardiovascular and cerebrovascular events (MACCE) by 25% in high-risk patients with established atherosclerotic cardiovascular disease (ASCVD) receiving guideline-directed medical therapy (GDMT). The consistent benefit across all components of the composite endpoint, coupled with manageable bleeding risk driven primarily by gastrointestinal events rather than intracranial hemorrhage, suggested a favorable benefit-risk profile in appropriately selected patients.

However, replication of cardiovascular outcomes trials in independent populations is critical for establishing treatment effects as robust and generalizable. Historical examples—including early fibrinolytic trials, glycoprotein IIb/IIIa inhibitor studies, and more recent investigations of novel antiplatelet strategies—have demonstrated that initial positive findings are not always reproduced in confirmatory studies, particularly when baseline risk profiles, background therapies, or event adjudication methods differ between trials.

The SHIELD-2 trial was designed as a confirmatory study to validate the MACCE reduction observed with cardioexaban in SHIELD-1, using identical eligibility criteria, treatment protocols, and endpoint definitions but enrolling an independent patient population at different sites with different investigators. Additionally, SHIELD-2 prospectively specified stroke as a key secondary endpoint, given the particularly robust 32% reduction observed in SHIELD-1 and the recognition that PAR-1 inhibition may have preferential effects on thrombotic stroke through mechanisms distinct from coronary thrombosis.

A unique aspect of SHIELD-2 was its conduct during a different phase of the COVID-19 pandemic compared with SHIELD-1. While SHIELD-1 enrolled primarily during the early pandemic period (March 2021 to November 2022), SHIELD-2 enrolled during a later phase (September 2022 to June 2023) when vaccination rates were higher and COVID-related healthcare disruptions had largely normalized. This provided an opportunity to assess whether the treatment effect of cardioexaban was robust across different baseline event rate environments.

2 METHODS

2.1 Trial Design and Oversight

SHIELD-2 was a multicenter, randomized, double-blind, placebo-controlled trial conducted at 387 sites in 26 countries. The trial design was nearly identical to SHIELD-1, with enrollment criteria, treatment protocols, and endpoint definitions harmonized to maximize comparability. The protocol was approved by institutional review boards or ethics committees at all sites. All patients provided written informed consent.

An independent data and safety monitoring board reviewed safety data at prespecified intervals. An independent clinical endpoint adjudication committee, blinded to treatment assignments, adjudicated all efficacy and safety endpoints using the same standardized definitions employed in SHIELD-1.

2.2 Patient Population

Eligibility criteria were identical to SHIELD-1. Patients were required to have documented ASCVD (prior MI >30 days, prior ischemic stroke >90 days, or symptomatic PAD) and elevated residual risk defined as either 10-year ASCVD risk score $\geq 15\%$ or ≥ 2 high-risk features (diabetes with target organ damage, CKD stage 3, or recurrent events within 24 months).

All patients were required to be on stable GDMT including antiplatelet therapy (aspirin 75-100 mg daily or clopidogrel 75 mg daily), statin therapy (high-intensity or maximally tolerated with LDL <100 mg/dL), and blood pressure control (systolic BP <140 mm Hg if tolerated). Patients with diabetes required HbA_{1c} <9.0%.

Exclusion criteria matched SHIELD-1: planned revascularization within 3 months, active bleeding,

history of intracranial hemorrhage, severe heart failure, severe renal impairment (eGFR <30 ml/min/1.73 m²), need for oral anticoagulation, or prior intolerance to P2Y₁₂ inhibitors.

2.3 Randomization and Treatment

Patients were randomly assigned 1:1 to cardioexaban 150 mg once daily or matching placebo using a central interactive web-response system. Randomization was stratified by geographic region, index qualifying event, and background antiplatelet therapy.

Trial visits occurred at 1 month, 3 months, and every 3 months thereafter. Medication adherence was assessed by pill counts. Background GDMT was continued and adjusted per standard of care.

2.4 Endpoints

The primary efficacy endpoint was time to first MACCE: cardiovascular death, myocardial infarction, stroke, or urgent coronary revascularization. All endpoint definitions were identical to SHIELD-1.

Key secondary endpoints, tested hierarchically, included: (1) three-point MACE (CV death, MI, or stroke); (2) stroke; (3) CV death; (4) myocardial infarction; (5) all-cause mortality.

The primary safety endpoint was TIMI major bleeding. Additional safety endpoints included TIMI minor bleeding, intracranial hemorrhage, gastrointestinal bleeding, and treatment discontinuation due to bleeding.

2.5 Statistical Analysis

SHIELD-2 was powered to detect a 20% relative risk reduction in MACCE with 90% power at alpha 0.05, requiring approximately 1,050 events. Sample size was calculated assuming a 5.5% annual event rate in the placebo group, similar to SHIELD-1.

The primary analysis used a Cox proportional-hazards model stratified by randomization factors in the intention-to-treat population. Cumulative incidence was estimated using Kaplan-Meier methods. Hierarchical testing controlled type I error for secondary endpoints.

A prespecified exploratory analysis pooled SHIELD-1 and SHIELD-2 data to provide more precise estimates of treatment effects, particularly for lower-frequency endpoints such as cardiovascular death and intracranial hemorrhage.

3 RESULTS

3.1 Patients

Between September 2022 and June 2023, 7,842 patients were enrolled and randomly assigned to cardioexaban (3,921 patients) or placebo (3,921 patients). Baseline characteristics were well balanced (Table 1) and similar to SHIELD-1. Mean age was 66.8 years, 29.4% were women, and 76.8% were white. The index qualifying event was prior MI in 56.8%, prior stroke in 25.9%, and PAD in 17.3%. Median 10-year ASCVD risk was 22.6% (IQR 17.3-30.1), with 65.1% having risk $\geq 20\%$. Diabetes was present in 63.4%, CKD in 42.1%, and prior coronary revascularization in 69.8%.

Median follow-up was 36.0 months (IQR 30.5-40.3). Adherence to trial medication at 36 months was 89.8% in the cardioexaban group and 90.4% in placebo. Permanent discontinuation occurred in 17.9% of cardioexaban patients (8.8% due to adverse events, 3.6% due to bleeding) and 13.8% of placebo patients (4.9% due to adverse events, 1.1% due to bleeding).

3.2 Primary Efficacy Endpoint

MACCE occurred in 449 patients (11.5%) in the cardioexaban group versus 590 patients (15.1%) in the placebo group (HR 0.73; 95% CI 0.63-0.85; $P < 0.001$) (Table 2, Figure 1). The absolute risk reduction was 3.6 percentage points, with a number needed to treat of 28 patients for 36 months.

Kaplan-Meier curves began separating at approximately 4 months and continued to diverge throughout follow-up. The annualized MACCE rate was 4.0% with cardioexaban versus 5.3% with placebo (rate ratio 0.75; 95% CI 0.67-0.84).

3.3 Secondary Efficacy Endpoints

Three-point MACE (CV death, MI, or stroke) occurred in 372 patients (9.5%) versus 503 patients (12.9%) (HR 0.71; 95% CI 0.59-0.85; $P < 0.001$). This 29% relative risk reduction was consistent with the SHIELD-1 finding (HR 0.77).

Individual components showed directionally consistent benefit:

- **Stroke:** 96 patients (2.5%) vs. 142 patients (3.6%) (HR 0.68; 95% CI 0.51-0.89; $P = 0.006$). This represented a 32% relative risk reduction, identical to SHIELD-1.
- **Cardiovascular death:** 134 patients (3.4%) vs. 169 patients (4.3%) (HR 0.79; 95% CI 0.61-1.03; $P = 0.08$).
- **Myocardial infarction:** 186 patients (4.7%) vs. 253 patients (6.5%) (HR 0.73; 95% CI 0.59-0.91; $P = 0.004$).
- **Urgent coronary revascularization:** 81 patients (2.1%) vs. 98 patients (2.5%) (HR 0.83; 95% CI 0.61-1.12; $P = 0.22$).

All-cause mortality occurred in 187 patients (4.8%) versus 227 patients (5.8%) (HR 0.82; 95% CI 0.67-1.01; $P = 0.06$).

3.4 Stroke Subtype Analysis

A prespecified exploratory analysis examined stroke subtypes. Among the 96 strokes in the cardioexaban group, 82 (85%) were ischemic, 9 (9%) were hemorrhagic, and 5 (5%) were undetermined. In the placebo group, 129 of 142 strokes (91%) were ischemic, 8 (6%) hemorrhagic, and 5 (4%) undetermined.

The reduction in ischemic stroke was significant (HR 0.64; 95% CI 0.47-0.86), while hemorrhagic stroke rates were numerically higher but not significantly different (HR 1.13; 95% CI 0.44-2.87; $P = 0.80$).

3.5 Safety Endpoints

TIMI major bleeding occurred in 160 patients (4.1%) in the cardioexaban group versus 90 patients (2.3%) in the placebo group (HR 1.77; 95% CI 1.35-2.33; $P<0.001$) (Table 3). The absolute increase was 1.8 percentage points (number needed to harm = 56).

Gastrointestinal bleeding accounted for 112 events (70%) in the cardioexaban group and 61 events (68%) in placebo. Intracranial hemorrhage occurred in 15 patients (0.4%) receiving cardioexaban and 11 patients (0.3%) receiving placebo (HR 1.36; 95% CI 0.63-2.95; $P=0.44$). Fatal bleeding occurred in 11 patients (0.3%) versus 7 patients (0.2%) ($P=0.48$).

TIMI minor bleeding occurred in 293 patients (7.5%) versus 172 patients (4.4%) (HR 1.71; 95% CI 1.40-2.08). Bleeding requiring transfusion occurred in 184 patients (4.7%) versus 102 patients (2.6%) (HR 1.80; 95% CI 1.41-2.30).

3.6 Pooled Analysis of SHIELD-1 and SHIELD-2

A prespecified pooled analysis of 16,048 patients from both trials showed:

- **MACCE:** HR 0.74 (95% CI 0.66-0.82; $P<0.001$)
- **Three-point MACE:** HR 0.74 (95% CI 0.65-0.84; $P<0.001$)
- **Stroke:** HR 0.68 (95% CI 0.56-0.83; $P<0.001$)
- **CV death:** HR 0.78 (95% CI 0.65-0.94; $P=0.009$)
- **TIMI major bleeding:** HR 1.80 (95% CI 1.48-2.18; $P<0.001$)
- **Intracranial hemorrhage:** HR 1.29 (95% CI 0.76-2.20; $P=0.35$)

The pooled analysis demonstrated cardiovascular death reduction reaching statistical significance with narrower confidence intervals, and confirmed that intracranial hemorrhage risk was not significantly elevated despite the larger sample size.

4 DISCUSSION

SHIELD-2 successfully confirmed the primary findings of SHIELD-1, demonstrating a 27% relative risk reduction in MACCE with cardioexaban in high-risk ASCVD patients receiving guideline-directed therapy. The consistency of results across two independent trials—conducted at different sites, during different time periods, with different investigators—provides strong evidence that the treatment effect is robust and reproducible.

Several aspects of the SHIELD-2 results merit emphasis. First, the magnitude of MACCE reduction (HR 0.73) was nearly identical to SHIELD-1 (HR 0.75), with overlapping confidence intervals, indicating highly consistent efficacy. The absolute risk reduction (3.6 percentage points vs. 3.8 percentage points) and number needed to treat (28 vs. 26) were also similar, supporting the generalizability of the benefit.

Second, the stroke reduction observed in SHIELD-2 (HR 0.68, 32% relative risk reduction) precisely replicated the SHIELD-1 finding. This consistency across trials is particularly noteworthy given that stroke was a prespecified key secondary endpoint in SHIELD-2 but not hierarchically tested in SHIELD-1. The stroke subtype analysis demonstrated that the benefit was driven by ischemic stroke prevention (HR 0.64), with no significant increase in hemorrhagic stroke despite the mechanism of action involving platelet inhibition.

The biologic rationale for cardioexaban's particularly robust effect on stroke relates to the distinct role of PAR-1 in cerebrovascular thrombosis. While both coronary and cerebral thrombosis involve platelet activation, *in vitro* studies suggest that thrombin-mediated platelet activation via PAR-1 plays a disproportionate role in cerebral microvascular thrombosis compared with coronary thrombosis, where ADP-mediated P2Y₁₂ activation may be more dominant. Dual-pathway inhibition appears to provide additive benefit in the cerebrovascular bed, consistent with the 32% stroke reduction observed in both SHIELD trials.

Third, the three-point MACE endpoint (excluding urgent revascularization) showed a 29% reduction in SHIELD-2, more pronounced than the 23% reduction observed in SHIELD-1. This suggests that the hard cardiovascular outcomes—death, MI, and stroke—may benefit more from dual-pathway inhibition than the softer endpoint of urgent revascularization, which is more susceptible to practice variation and adjudication differences across sites.

The safety profile in SHIELD-2 closely mirrored SHIELD-1. TIMI major bleeding was increased by 1.8 percentage points (HR 1.77), with 70% of events being gastrointestinal. Critically, intracranial hemorrhage rates remained low and balanced between groups (0.4% vs. 0.3%), and the pooled analysis of 16,048 patients confirmed no significant ICH signal (HR 1.29; 95% CI 0.76-2.20). This finding is particularly reassuring given that intracranial hemorrhage is the most feared complication of antiplatelet therapy and the endpoint that most strongly influences clinical decision-making.

The net clinical benefit—balancing MACCE reduction against bleeding risk—favored cardioexaban in both trials. In SHIELD-2, for every 56 patients treated for 36 months, one major bleeding event was caused, but for every 28 patients treated, one MACCE event was prevented. This 2:1 benefit-to-harm ratio is consistent with SHIELD-1 and supports the use of cardioexaban in high-risk populations where the baseline MACCE risk justifies the bleeding trade-off.

The pooled analysis of SHIELD-1 and SHIELD-2 provided several important insights. First, the cardiovascular death reduction reached statistical significance (HR 0.78; 95% CI 0.65-0.94; $P=0.009$), an outcome that was directionally favorable but not significant in either trial individually due to lower event rates. Second, the stroke benefit was highly robust (HR 0.68; 95% CI 0.56-0.83), with the upper bound of the confidence interval excluding no effect. Third, the pooled intracranial hemorrhage analysis definitively demonstrated no significant increase in this catastrophic outcome, despite doubling the sample size and improving precision.

One notable difference between SHIELD-1 and SHIELD-2 relates to the pandemic environment. SHIELD-1 enrolled during the early COVID-19 period when social distancing, reduced healthcare utilization, and

lower community respiratory infection rates may have suppressed cardiovascular event rates. SHIELD-2 enrolled during a later phase when vaccination was widespread and healthcare systems had normalized. Despite these different baseline environments, the treatment effect was consistent, suggesting that cardioexaban's benefit is not contingent on specific background event rate conditions.

The trial has limitations. The population was predominantly white and male, though racial and gender representation was similar to other contemporary cardiovascular outcomes trials. Background antiplatelet therapy was limited to aspirin or clopidogrel; efficacy and safety with ticagrelor or prasugrel were not evaluated. Patients with severe renal impairment, need for oral anticoagulation, and recent acute coronary syndrome were excluded. Median follow-up of 36 months may not capture very long-term outcomes.

In conclusion, SHIELD-2 confirmed that cardioexaban reduces MACCE in high-risk ASCVD patients on guideline-directed therapy, with particularly robust stroke prevention and an acceptable bleeding profile characterized by gastrointestinal events but not intracranial hemorrhage. The replication of findings across two large, independent trials provides strong evidence supporting the use of cardioexaban in appropriately selected high-risk secondary prevention populations.

Table 1: Baseline Characteristics of the Patients

Characteristic	Cardioexaban (N=3921)	Placebo (N=3921)
Age — yr	66.7 ± 9.9	66.9 ± 9.8
Female sex — no. (%)	1153 (29.4)	1151 (29.4)
Race — no. (%)		
White	3012 (76.8)	3009 (76.7)
Black	405 (10.3)	410 (10.5)
Asian	342 (8.7)	338 (8.6)
Body-mass index — kg/m ²	29.6 ± 5.3	29.8 ± 5.5
Index qualifying ASCVD event — no. (%)		
Prior myocardial infarction	2227 (56.8)	2223 (56.7)
Prior ischemic stroke	1015 (25.9)	1017 (25.9)
Peripheral arterial disease	679 (17.3)	681 (17.4)
Cardiovascular risk factors — no. (%)		
Diabetes mellitus	2486 (63.4)	2481 (63.3)
Chronic kidney disease (eGFR 30-59)	1651 (42.1)	1648 (42.0)
Hypertension	3521 (89.8)	3518 (89.7)
10-year ASCVD risk score		
Median (IQR) — %	22.7 (17.4-30.2)	22.5 (17.2-30.0)
≥20% — no. (%)	2553 (65.1)	2549 (65.0)
Medications at baseline — no. (%)		
Aspirin	2538 (64.7)	2541 (64.8)
Clopidogrel	1383 (35.3)	1380 (35.2)
Statin therapy	3865 (98.6)	3863 (98.5)
High-intensity statin	3230 (82.4)	3225 (82.2)
Laboratory values at baseline		
LDL cholesterol — mg/dL	77.8 ± 27.9	78.1 ± 28.2
eGFR — ml/min/1.73 m ²	68.2 ± 18.4	68.5 ± 18.6

Table 2: Efficacy Endpoints

Endpoint	Cardioexaban (N=3921)	Placebo (N=3921)	Hazard Ratio (95% CI)	P Value
Primary endpoint				
MACCE (4-point)	449 (11.5)	590 (15.1)	0.73 (0.63-0.85)	<0.001
Key secondary endpoints				
MACE (3-point)	372 (9.5)	503 (12.9)	0.71 (0.59-0.85)	<0.001
Stroke	96 (2.5)	142 (3.6)	0.68 (0.51-0.89)	0.006
Ischemic stroke	82 (2.1)	129 (3.3)	0.64 (0.47-0.86)	0.003
Hemorrhagic stroke	9 (0.2)	8 (0.2)	1.13 (0.44-2.87)	0.80
CV death	134 (3.4)	169 (4.3)	0.79 (0.61-1.03)	0.08
Myocardial infarction	186 (4.7)	253 (6.5)	0.73 (0.59-0.91)	0.004
Urgent revascularization	81 (2.1)	98 (2.5)	0.83 (0.61-1.12)	0.22
All-cause mortality	187 (4.8)	227 (5.8)	0.82 (0.67-1.01)	0.06

Table 3: Safety Endpoints

Endpoint	Cardioexaban (N=3921)	Placebo (N=3921)	Hazard Ratio (95% CI)	P Value
Primary safety endpoint				
TIMI major bleeding	160 (4.1)	90 (2.3)	1.77 (1.35-2.33)	<0.001
Components				
Intracranial hemorrhage	15 (0.4)	11 (0.3)	1.36 (0.63-2.95)	0.44
GI bleeding	112 (2.9)	61 (1.6)	1.83 (1.33-2.52)	<0.001
Fatal bleeding	11 (0.3)	7 (0.2)	1.57 (0.61-4.07)	0.48
Other endpoints				
TIMI minor bleeding	293 (7.5)	172 (4.4)	1.71 (1.40-2.08)	<0.001
Bleeding requiring transfusion	184 (4.7)	102 (2.6)	1.80 (1.41-2.30)	<0.001

Table 4: Pooled Analysis of SHIELD-1 and SHIELD-2

Endpoint	Cardioexaban (N=8024)	Placebo (N=8024)	Hazard Ratio (95% CI)	P Value
Efficacy				
MACCE (4-point)	938 (11.7)	1233 (15.4)	0.74 (0.66-0.82)	<0.001
MACE (3-point)	784 (9.8)	1034 (12.9)	0.74 (0.65-0.84)	<0.001
Stroke	204 (2.5)	299 (3.7)	0.68 (0.56-0.83)	<0.001
CV death	276 (3.4)	350 (4.4)	0.78 (0.65-0.94)	0.009
Myocardial infarction	384 (4.8)	521 (6.5)	0.73 (0.63-0.85)	<0.001
Safety				
TIMI major bleeding	316 (3.9)	176 (2.2)	1.80 (1.48-2.18)	<0.001
Intracranial hemorrhage	31 (0.4)	24 (0.3)	1.29 (0.76-2.20)	0.35
GI bleeding	221 (2.8)	119 (1.5)	1.86 (1.48-2.34)	<0.001

DISCLAIMER: This is a fictional clinical trial created for educational and training purposes only. Cardioexaban is not a real drug. Novellia Pharmaceuticals is a fictional entity. All data, patient characteristics, and trial results are simulated for the SHIELD Socratic AI Training System demonstration. This document should not be used for clinical decision-making or regulatory purposes.